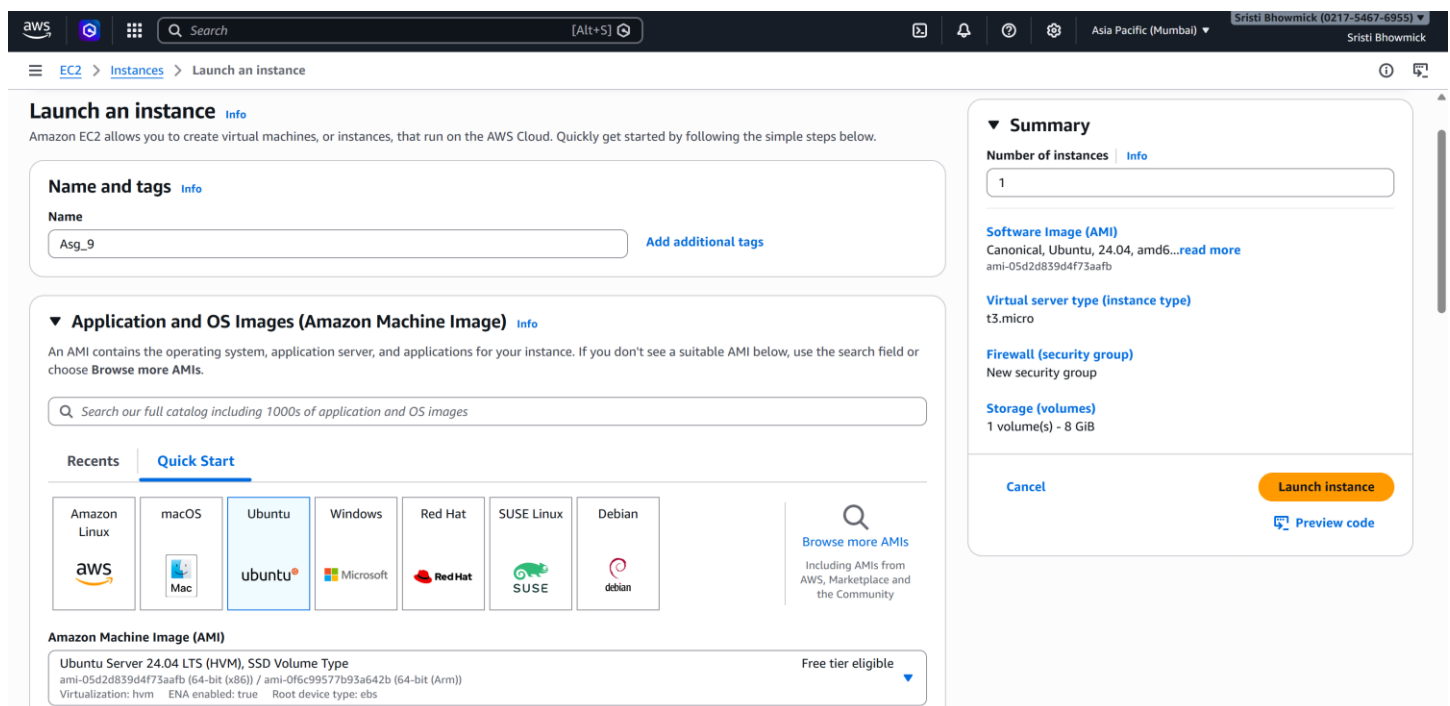
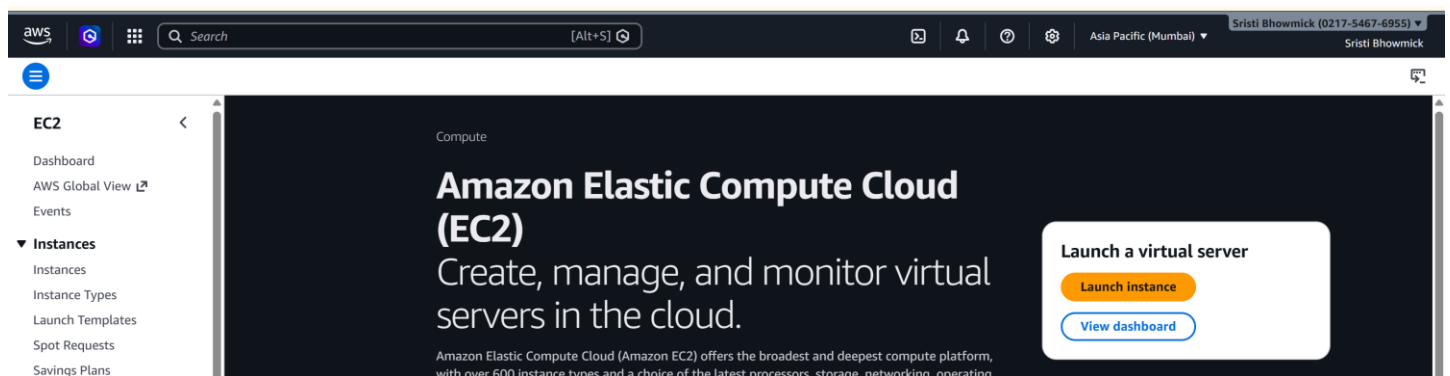
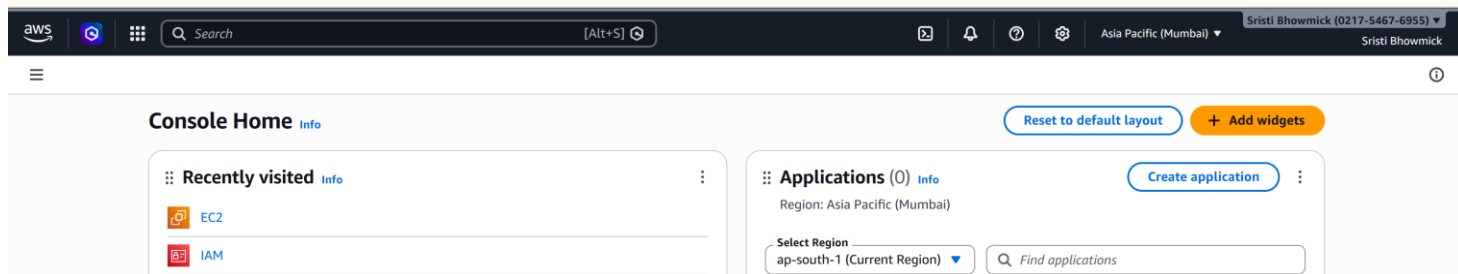


NAME : SRISTI BHOWMICK , COLLEGE ROLL : BTECH/CSE/23/095 , EXAM ROLL : 11600223120

Assignment 9: Deploy a project from GitHub to EC2.

Steps: login to aws console -> EC2 -> Launch an instance -> select ubuntu -> create key pair -> open bitwise -> add key.pem in client key manager -> login -> open terminal.



Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

RSA encrypted private and public key pair

☐ ED25519

ED25519 encrypted private and public key pair

Private key file format

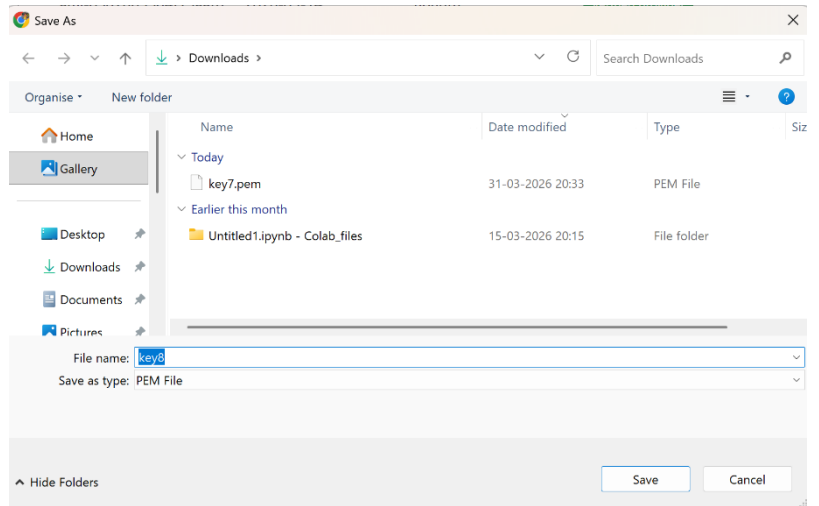
☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

[Cancel](#)[Create key pair](#)

[EC2](#) > [Instances](#) > Launch an instance

Info Help

Network settings

[Edit](#)

Network

vpc-0a874208a064ece04

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group☐ Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05d2d839d4f73aafb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)[Launch instance](#)[Preview code](#)

[EC2](#) > [Instances](#) > Launch an instance

Info Help

Success

Successfully initiated launch of instance (i-041afe227112c6d69)



[EC2](#) > [Instances](#) > i-041afe227112c6d69

Info Help

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Instance summary for i-041afe227112c6d69 (Asg_9)

Updated less than a minute ago

Instance ID

i-041afe227112c6d69

IPv6 address

-

Public IPv4 address

3.109.49.197 | [open address](#)

Instance state

Running

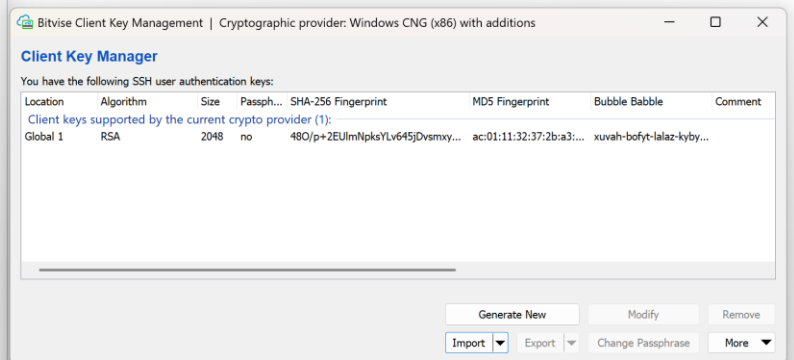
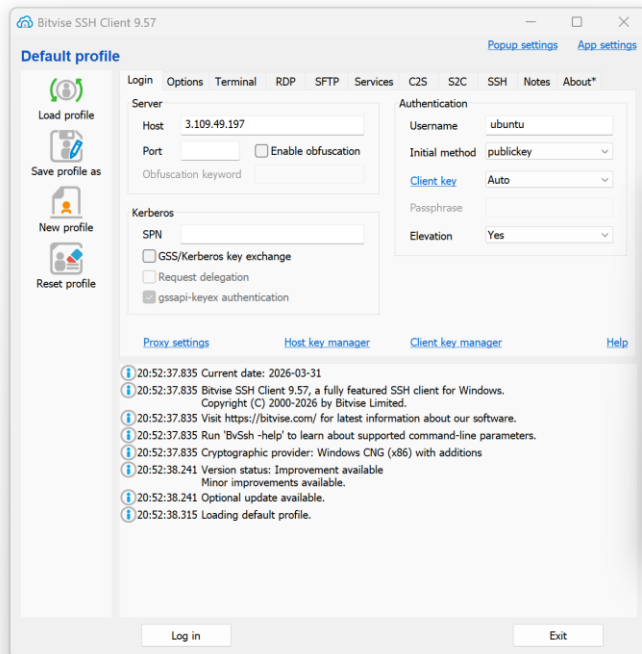
[Connect](#)[Instance state](#)[Actions](#)

Private IPv4 addresses

172.31.32.203

Public DNS

ec2-3-109-49-197.ap-south-1.compute.amazonaws.com | [open address](#)



Step2: run commands

1. `sudo apt-get update`
2. `sudo apt-get upgrade`
3. `sudo apt-get install nginx`
4. `curl -SL https://deb.nodesource.com/setup_16.x|sudo -E bash -`
5. `sudo apt install nodejs`

```
ubuntu@ip-172-31-32-203:~$ sudo apt-get update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
```

```
ubuntu@ip-172-31-32-203:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
```

```
ubuntu@ip-172-31-32-203:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
```

```
ubuntu@ip-172-31-32-203:~$ curl -SL https://deb.nodesource.com/setup_16.x|sudo -E bash -
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload    Total   Spent    Left   Speed
100  5736  100  5736    0     0  101k      0  --:--:-- --:--:-- --:--:-- 103k
```

```
ubuntu@ip-172-31-32-203:~$ sudo apt install nodejs
Reading package lists... Done
```

Step3: clone github project repository by git clone -> start server by node index.js

```
ubuntu@ip-172-31-32-203:~$ git clone https://github.com/bhowmicksristi29-beep/Repository2.git
Cloning into 'Repository2'...
```

```
ubuntu@ip-172-31-32-203:~$ cd Repository2
```

```
ubuntu@ip-172-31-32-203:~/Repository2$ npm install
```

```
ubuntu@ip-172-31-32-203:~/Repository2$ node index.js
Started server
```

← → ↻ ⚠ Not secure 3.109.49.197

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 4: change security rule -> security-> edit -> add rule -> add a rule with ipv4 and port=4000 - > save -> and refresh the page of ip site with port = 4000 -> our project will be displayed

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

-

Security groups

[sg-0016d9776db27d899 \(launch-wizard-2\)](#)

Owner ID

[021754676955](#)

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

🔄

Manage tags

Edit inbound rules

🔍 Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-0f61f9975c49e32a8	IPv4	HTTP	TCP	80	0.0.0.0/0
☐	-	sgr-01769f42c1de55fbd	IPv4	SSH	TCP	22	0.0.0.0/0
☐	-	sgr-0433a780e8f1100cc	IPv4	HTTPS	TCP	443	0.0.0.0/0

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Sristi Bhowmick (0217-5467-6955)

EC2

Security Groups

sg-0016d9776db27d899 - launch-wizard-2

Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID

sg-0f61f9975c49e32a8

Type

HTTP

Protocol

TCP

Port range

80

Source

Custom

Q

0.0.0.0/0

Delete

sg-01769f42c1de55fbd

SSH

TCP

22

Custom

Q

0.0.0.0/0

Delete

sg-0433a780e8f1100cc

HTTPS

TCP

443

Custom

Q

0.0.0.0/0

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

EC2

Security Groups

sg-0016d9776db27d899 - launch-wizard-2

Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID

sg-0f61f9975c49e32a8

Type

HTTP

Protocol

TCP

Port range

80

Source

Custom

Q

0.0.0.0/0

Delete

sg-01769f42c1de55fbd

SSH

TCP

22

Custom

Q

0.0.0.0/0

Delete

sg-0433a780e8f1100cc

HTTPS

TCP

443

Custom

Q

0.0.0.0/0

Delete

-

Custom TCP

TCP

4000

Anywh...

Q

0.0.0.0/0

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

←

→

↻

⚠ Not secure

3.109.49.197:4000

Hello World